



Accelerating Business Growth and Maximizing Telecom Revenue Using AI.

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Current State of the Telecom Retention Industry.

In Telecom Churn, customers migrate when companies fail the **4 C's**:

Cost

Uncompetitive pricing plans or sudden billing spikes.

Coverage

Poor network quality, dropped calls or slow data speeds.

Customer Service

Frustrating, slow or unresolved support interactions.

Convenience

Rigid contracts and friction during plan upgrades.

The ROI Angle: Retaining a subscriber protects our baseline revenue and eliminates high Customer Acquisition Costs (CAC).

From large volumes of information such as customer data and behaviour, it is now possible to represent complex relationships that humans could not discover.



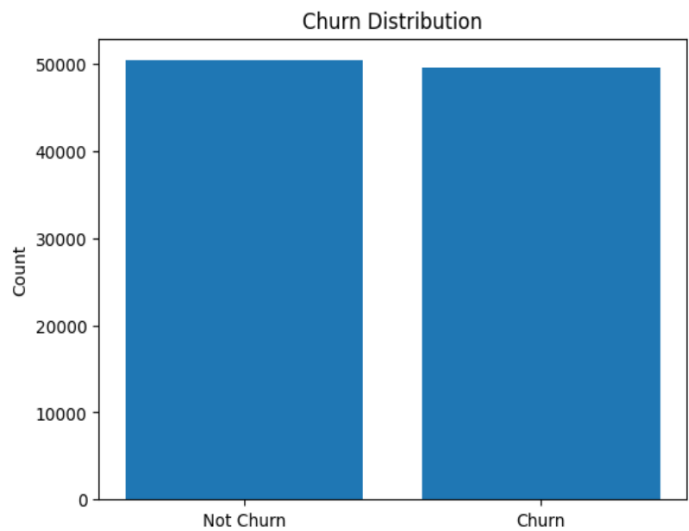
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Prediction using Machine Learning Models.

Over **100,000 distinct historical customer profiles** have been accumulated as behavioral data, tracking **100 unique performance columns** to monitor every angle of subscriber activity.

unqsubs	actvsubs	new_cell	crclscod	asl_flag	totcalls	totmou	totrev	adjrev	adjmou		
2	1	U	A	N	1652	4228.00000	1504.62	1453.44	4085.00		
opk_dat_Mean	mou_opkv_Mean	mou_opkd_Mean	drop_blk_Mean	attempt_Mean	complete_Mean	callfwdv_Mean	callwait_Mean	churn	months		
0.0	55.220000	0.0	1.333333	52.333333	45.000000	0.0	0.333333	1	61		
0.0	169.343333	0.0	9.333333	263.333333	193.333333	0.0	5.666667	0	56		
0.0	0.233333	0.0	0.333333	9.000000	6.000000	0.0	0.000000	1	58		
0.0	5.450000	0.0	0.000000	3.666667	3.666667	0.0	0.000000	0	60		
0.0	218.086667	0.0	10.333333	222.333333	137.000000	0.0	0.000000	0	57		



The dataset was relatively balanced between churn and non-churn customers, making Accuracy a meaningful evaluation metric. However, ROC-AUC was selected as the primary optimization metric because it evaluates the model's ability to distinguish between churners and non-churners across all possible decision thresholds. This provides a more robust assessment of customer ranking and supports future business decisions where different retention strategies may require different thresholds.

Data Scope & Information Sources



Customer Demographics & Household Information

- Gender and marital status
- Household size and presence of children
- Residential characteristics (length of residence, dwelling type, ownership status)
- Socioeconomic indicators (income level, credit class, social segmentation)

Service Usage Behavior

- Monthly and historical voice/data usage (minutes of use, call counts)
- Call types (completed, dropped, blocked, unanswered, roaming)
- Peak vs off-peak usage distribution
- Customer service interactions and support calls

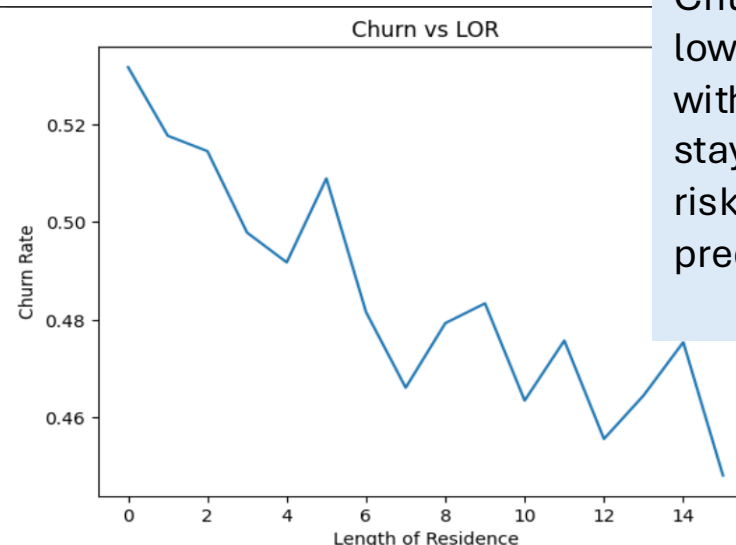
Financial & Billing Information

- Monthly revenue and total lifetime revenue
- Overage charges (voice and data)
- Average and adjusted billing metrics
- Equipment pricing and handset cost information
- Price sensitivity indicators (revenue per usage unit)

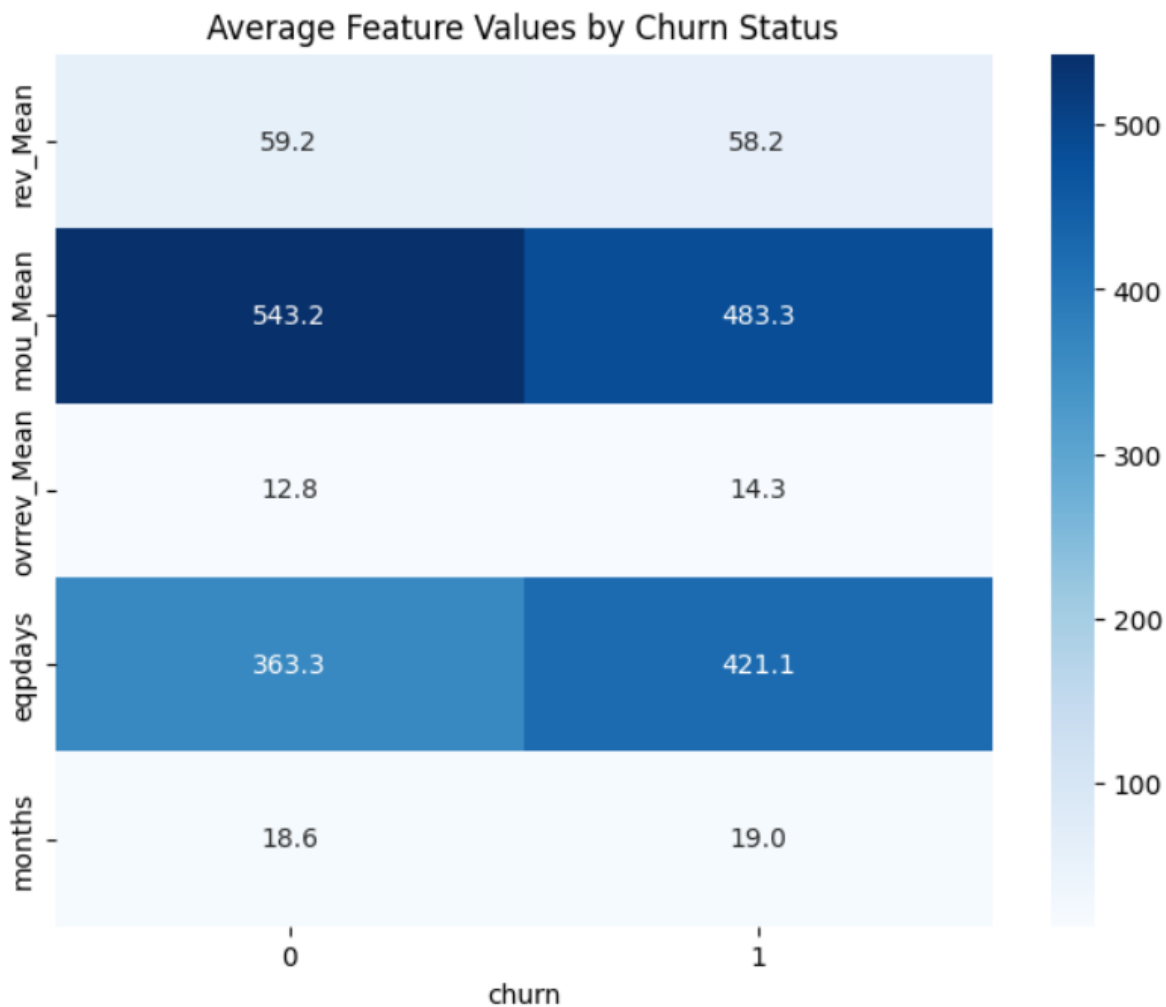
Target Variable

- Customer churn status within a defined observation window (31–60 days)

Data Visualization



Churn is higher among customers with older devices and lower length of residence. Customers who have been with the company for a longer period are more likely to stay, while ageing equipment appears to increase churn risk. Both factors provide valuable signals for churn prediction and retention strategies.

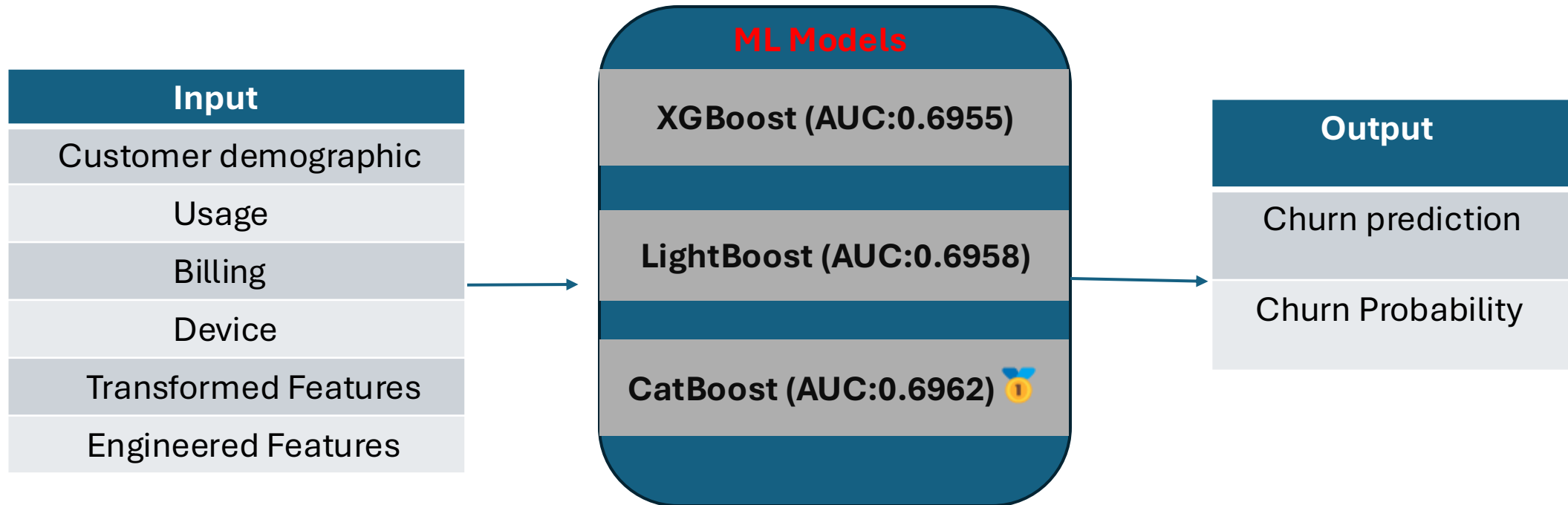


Data Visualization

- Household Subscribers (uniqsubs): Churn rate generally increased as the number of subscribers within the same household increased, suggesting that larger accounts may exhibit higher churn risk.
- Revenue Distribution (rev_Mean): Revenue showed a highly right-skewed distribution, with most customers concentrated in lower revenue ranges. Revenue patterns were relatively similar between churners and non-churners.
- Usage Behavior (mou_Mean): Non-churning customers tended to have higher average monthly usage, indicating that more engaged customers are less likely to leave the service.
- Overage Revenue (ovrrrev_Mean): Customers who churned generated slightly higher overage charges on average, suggesting that unexpected or excessive usage costs may contribute to customer dissatisfaction.
- Equipment Age (eqpdays): Customers who churned had noticeably older devices than retained customers. This indicates that aging equipment may be associated with increased churn risk and represents a potential retention opportunity.
- Customer Tenure (months / LOR): An inverse relationship was observed between customer tenure and churn. Customers with shorter tenure exhibited higher churn rates, while long-term customers demonstrated stronger retention.

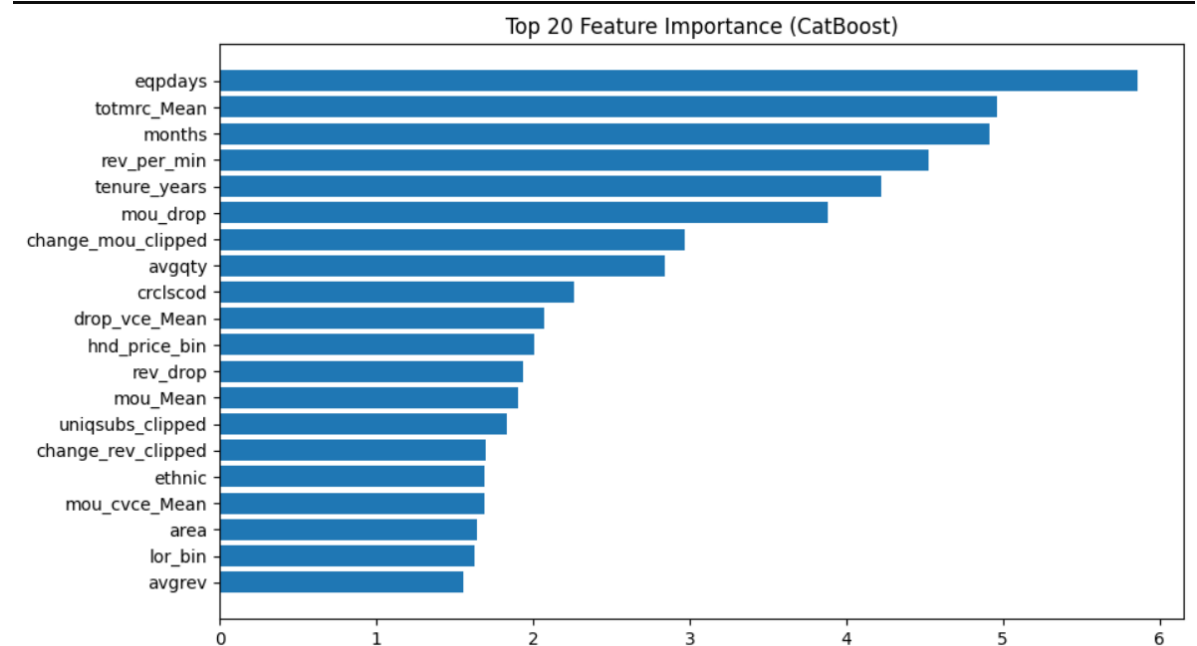
Overview of the Machine Learning Model

Three gradient boosting models were evaluated: XGBoost, LightGBM and CatBoost. These were chosen for their strong performance on tabular data and ability to capture complex non-linear patterns in churn behavior. Since the main business goal is customer retention, Recall was prioritized to better capture potential churners early and enable timely intervention. In addition, ROC-AUC and F1-Score were used to assess ranking quality and overall balance between precision and recall.



Important Features for Prediction

- Features derived from processed data and statistical measures are important for AI predictions, indicating that the AI takes a different approach from human decision-making.



Customer Risk Segmentation & Decile Analysis

- Customers were segmented based on predicted churn probability into Low, Medium and High-Risk groups. The High-Risk segment(**28.4%**) represents customers who should be prioritized for retention campaigns, while Medium-Risk(**58.2%**)customers can be targeted with preventive actions and Low-Risk(**13.4%**) customers require regular engagement only.
- Decile Analysis and Lift were used to assess the model's ability to rank customers by churn risk. The top 10% highest-risk customers achieved a churn rate of 78.8%, compared to an overall churn rate of 49.6%, yielding a Lift of 1.59. This indicates that the model successfully identifies customers with a significantly higher likelihood of churn and can support targeted retention strategies.

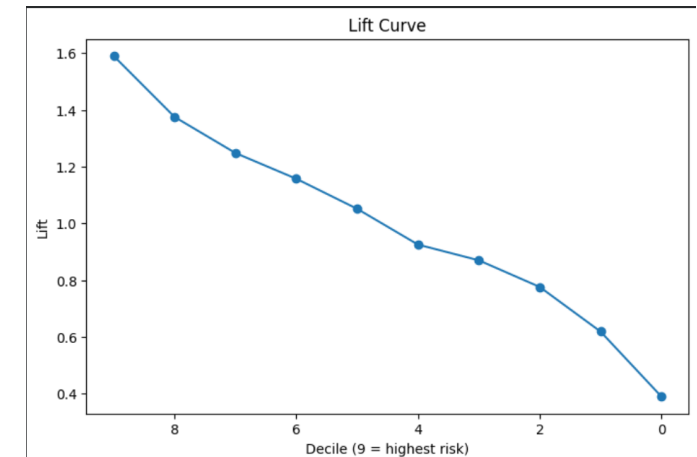
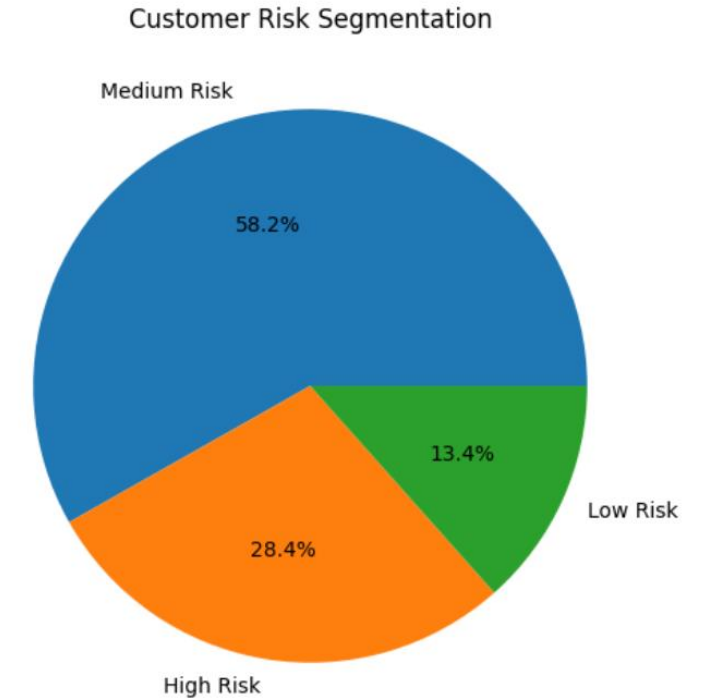


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Our Proactive Customer Retention Proposal

eqpdays (Equipment Days)

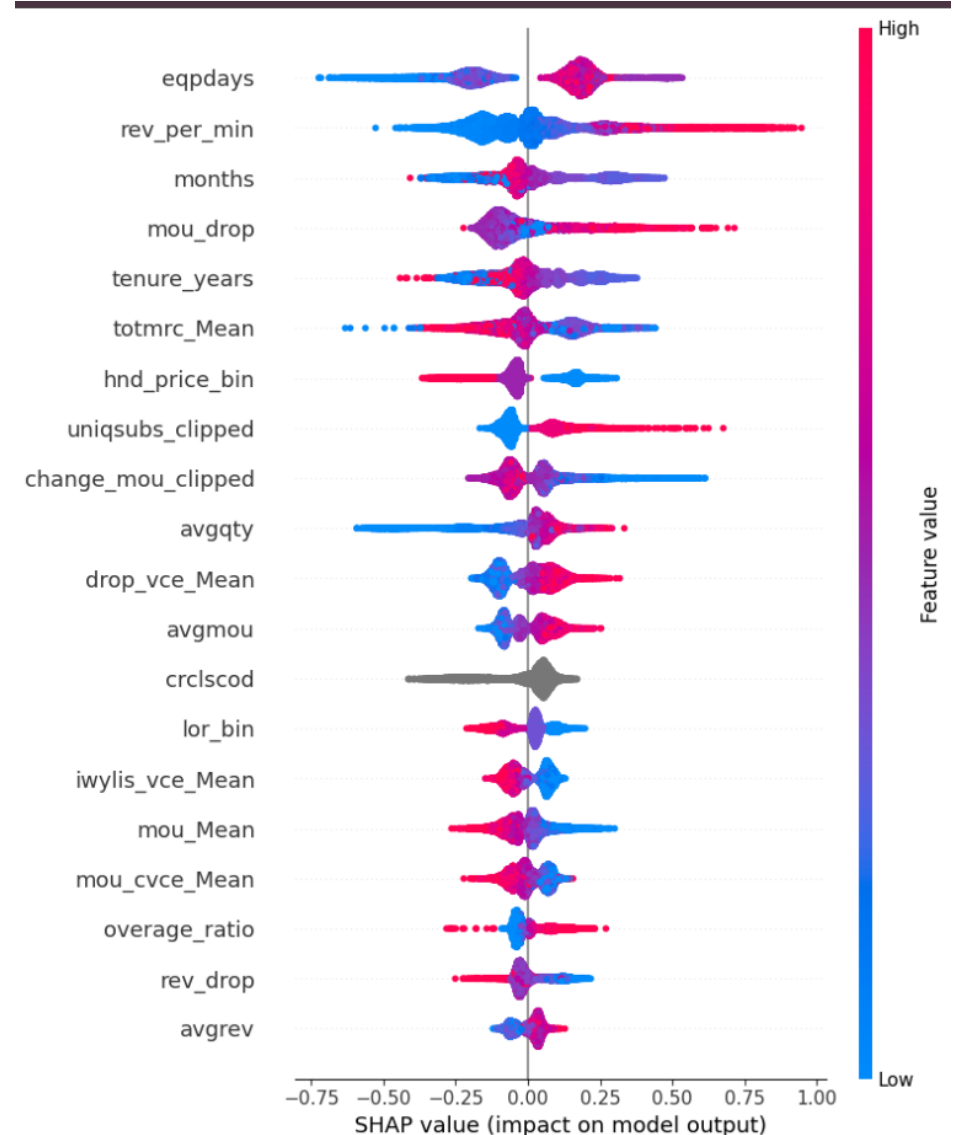
- Customers who have been using the same handset for a long time (older phones) have a significantly higher probability of churning. This indicates a strong correlation between device obsolescence and customer attrition.
- Actionable Recommendation:** Launch a targeted **"Device Upgrade Program."** Proactively approach these customers with trade-in offers, installment plans, or discounts on new smartphones to lock them into new contract cycles.

rev_per_min (Revenue per Minute of Usage)

- Customers who experience a high cost per minute (spending a lot of money relative to few minutes of actual use) are highly prone to churning. They likely perceive their current rate plans as unfair, overpriced, or poor value for money.
- Actionable Recommendation:** Implement an automated **"Plan Optimizer"** system. Before these customers switch to competitors, proactively recommend tailored, more cost-effective voice/data bundles that offer better value for their actual consumption patterns

months & tenure_years (Customer Account Tenure)

- New customers (low tenure) represent the highest risk of immediate churn. Once a customer survives the critical initial period, their loyalty and stability increase significantly.
- Actionable Recommendation:** Strengthen the **"Onboarding and First-90-Days Experience."** Enhance customer care touchpoints during the first 3 to 6 months, ensuring high service quality and proactive satisfaction checks during this vulnerable phase.



Our Proactive Customer Retention Proposal

mou_drop (Drop in Minutes of Usage)

- A sharp decline in a customer's talk time compared to their historical 3-month average is a powerful **Early Warning Indicator** of behavioral churn. The customer is already disengaging from the network.
- **Actionable Recommendation:** Deploy an automated "**Usage Trigger Alert System.**" The moment a customer's monthly usage drops below a specific percentage threshold, trigger an immediate retention workflow (e.g., automated bonus credits or a customer care courtesy call to address underlying issues).

hnd_price_bin (Handset Price Tier / Bin)

- Premium subscribers with expensive, high-end smartphones (Red) exhibit strong network retention. Conversely, budget-conscious users with economic or low-end devices (Blue) are highly volatile and easily tempted by competitors' promotional pricing.
- **Actionable Recommendation:** Create a "**Budget-Tier Loyalty Campaign.**" Offer low-cost, high-incentive value perks (such as free social media data or family plan discounts) specifically designed to retain price-sensitive segments.

